

Practice Quiz: Action

Part A: Multiple Choice

1. The action equation $da/dt = -\partial F/\partial a$ describes: A) How beliefs are updated B) How actions change sensory states to minimize free energy — "making predictions come true" C) How parameters are learned D) Static equilibrium only
2. Expected Free Energy $G(\pi)$ differs from variational free energy F in that: A) G evaluates *future* outcomes under a policy, while F evaluates *current* observations B) G is always smaller than F C) G does not involve probabilities D) G is used only for perception
3. The pragmatic value component of EFE corresponds to: A) Information gain B) How well expected outcomes align with preferred outcomes (the C vector) C) The entropy of the prior D) The precision of observations
4. The epistemic value component of EFE corresponds to: A) Reward maximization B) Expected information gain — how much a policy will reduce uncertainty about hidden states C) The log-likelihood D) The number of actions available
5. In the softmax policy selection $P(\pi) = \sigma(-\gamma \cdot G(\pi))$, the inverse temperature γ controls: A) The speed of action execution B) The precision of policy selection — high $\gamma \rightarrow$ more deterministic; low $\gamma \rightarrow$ more random C) The number of policies considered D) The prior probability of each policy
6. The A matrix in a POMDP encodes: A) Actions B) The likelihood mapping $p(o|s)$ — how hidden states generate observations C) Transition dynamics D) Preferences
7. The C vector in a POMDP encodes: A) Transition probabilities B) Log-prior preferences over observations — $\ln p(o)$ C) The likelihood D) The initial state distribution

Part B: Short Answer

1. For a two-policy POMDP with $G(\pi_1) = -2$ and $G(\pi_2) = -5$, compute $P(\pi_1)$ and $P(\pi_2)$ for $\gamma = 1$ and $\gamma = 4$. Show how increasing γ shifts probability toward the lower-EFE policy.
2. Explain the mathematical relationship between Expected Free Energy and the KL divergence between predicted and preferred outcomes. How does this formalize "goal-directed behavior"?
3. A policy π_1 has high pragmatic value but low epistemic value. Policy π_2 has low pragmatic value but high epistemic value. Under what conditions would the agent prefer π_2 ? How does this relate to curiosity-driven behavior?